

DECK 08 · WEEK 4 · MORNING 1

# FSA III — Taxes, Liabilities & Quality

*The analyst toolkit that turns a set of statements into a view on the business*

CFA Level I · Vol 4 Learning Modules 8–12

# Roadmap — FSA III — Taxes, Liabilities, Quality, Ratios & Modeling

6 sections · ~2 hours · Built bottom-up from CFA readings

01

## Long-term liabilities & leases

Five finance-lease criteria · ROU asset + lease liability · Proton Enterprises finance vs operating · lessor accounting

02

## Pensions & share-based comp

DC vs DB · rolling the defined-benefit obligation forward · IFRS vs US GAAP · grants and options at grant-date fair value

03

## Income taxes

Accounting profit vs taxable income · DTL / DTA direction · Reston Partners · statutory, effective and cash rates at J&J

04

## Reporting quality & red flags

Reporting vs earnings quality · the six-step spectrum · fraud triangle · the warning-sign battery · CFO to net income

05

## Ratio analysis & DuPont

Activity · liquidity · solvency · profitability · Anson Industries three-way and Amsterdam PLC five-way decomposition

06

## 3-statement modeling

The Rémy Cointreau build sequence · working capital in days · scenario analysis as the cure for overconfidence

# Learning objectives

*By the end of this deck you should be able to:*

1. Classify a lease against the five criteria, capitalise it as a right-of-use asset and lease liability, and trace the finance-versus-operating split through the income statement, cash flows and ratios.
2. Distinguish defined-contribution from defined-benefit plans, roll a defined-benefit obligation forward, and split the movement between profit or loss and other comprehensive income.
3. Separate permanent from temporary differences, decide whether a difference creates a deferred tax asset or a liability, and contrast the statutory, effective and cash tax rates.
4. Place a set of statements on the reporting-quality spectrum, apply the fraud triangle, and run the curriculum warning-sign checks including cash from operations over net income.
5. Compute activity, liquidity, solvency and profitability ratios, and decompose return on equity three ways and five ways.
6. Build a three-statement forecast in the curriculum sequence, drive working capital off inventory, receivable and payable days, and bracket free cash flow with scenarios.

SECTION 01

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# Long-term liabilities & leases

# Long-term liabilities come in four flavours – bonds, leases, pensions and deferred tax – and only one of them is labelled debt on the face of the balance sheet

*Reported "total debt" is rarely economic debt. The footnotes carry the rest, and the analyst has to put every peer on one yardstick.*

| Instrument    | Where it sits             | Cash nature            | Analyst adjustment             |
|---------------|---------------------------|------------------------|--------------------------------|
| Bonds payable | Amortised cost on the BS  | Coupons + principal    | Mark to market if yields moved |
| Leases        | ROU asset + lease liab.   | Fixed periodic payment | Include in debt at PV          |
| Pensions (DB) | Net funded status: A or L | Future benefit payouts | Add net deficit to debt        |
| Deferred tax  | DTL / DTA line            | No cash until reversal | Not debt, but flag reversals   |

## THE ECONOMIC-DEBT LENS

Economic debt = bonds + capitalised leases + net pension deficit – surplus cash. Before IFRS 16 and ASC 842 pulled leases onto the balance sheet, reported debt understated true leverage at every airline, retailer and restaurant chain. The standards closed that gap; pensions and deferred tax are still footnote work.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Introduction" and Learning Module Overview, pp. 245–247.*

# A lease is a finance lease if ANY one of five criteria is met — but either way it lands on the balance sheet as a right-of-use asset and a lease liability

*Two exemptions survive: a term of 12 months or less (IFRS and US GAAP), and low-value assets up to about USD 5,000 in sales price (IFRS only).*

$$\text{ROU asset} = \text{Lease liability} = \sum \text{Payment}_t / (1 + r)^t$$

*where:  $r$  = the rate implicit in the lease, or the lessee's incremental borrowing rate. The same five criteria are used by lessees and lessors, and are identical under IFRS and US GAAP.*

## FIVE FINANCE-LEASE CRITERIA · ANY ONE TRIGGERS

1. Ownership of the asset transfers to the lessee
2. A purchase option the lessee is reasonably certain to take
3. The term covers a major part of the asset's useful life
4. PV of payments  $\geq$  substantially all of the asset's fair value
5. The asset has no alternative use to the lessor

## IFRS VS US GAAP IN ONE LINE

The balance sheet is identical under both. Only the lessee income statement differs. IFRS runs a SINGLE model — every lease reports separate interest and amortisation, like a finance lease. US GAAP keeps a second model for operating leases, where the lessee shows one flat "lease expense" inside operating costs and puts the whole payment through operating cash flow.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Lease Classification as Finance or Operating" and Example 1, pp. 247–250.*

# Same cash and same total expense — the finance lease front-loads the charge while the US GAAP operating lease delivers a flat one

Five years, EUR 100,000 a year in arrears, 10% implicit rate. PV = EUR 379,079 — the opening ROU asset and lease liability under both classifications.

| Year         | Finance: interest | Finance: amort. | Finance: total | Operating: flat | Gap (F – O) |
|--------------|-------------------|-----------------|----------------|-----------------|-------------|
| 1            | 37,908            | 75,816          | 113,724        | 100,000         | +13,724     |
| 2            | 31,699            | 75,816          | 107,515        | 100,000         | +7,515      |
| 3            | 24,869            | 75,816          | 100,685        | 100,000         | +685        |
| 4            | 17,355            | 75,816          | 93,171         | 100,000         | –6,829      |
| 5            | 9,091             | 75,816          | 84,907         | 100,000         | –15,093     |
| <b>Total</b> | <b>120,921</b>    | <b>379,079</b>  | <b>500,000</b> | <b>500,000</b>  | <b>0</b>    |

## WHY THE FINANCE LEASE HURTS EARLY

Amortisation is flat at 75,816 every year. Interest is charged on a shrinking liability, so it starts at 37,908 and ends at 9,091. Flat plus falling beats a flat 100,000 early and loses to it late. Cash out is EUR 100,000 a year under both, and the five-year totals are identical — nothing economic has changed.

Source: CFA Institute, Curriculum Vol 4, Learning Module 8, Examples 2 and 3 (Proton Enterprises), pp. 251–254.

# Using the operating-lease model lowers EBITDA margin, asset turnover AND operating cash flow at once – so peers on different models are not comparable

*The curriculum asks the question one way round: what happens if you use an operating lease INSTEAD of a finance lease? Almost every answer is "lower".*

| Ratio               | Formula                  | Operating vs finance | Why                                |
|---------------------|--------------------------|----------------------|------------------------------------|
| EBITDA margin       | EBITDA / Revenue         | Lower                | Lease is op-ex, not below EBITDA   |
| Operating margin    | Operating income / Rev   | Lower                | Flat 100,000 vs 75,816 amort.      |
| Asset turnover      | Revenue / Avg assets     | Lower                | ROU asset amortises more slowly    |
| Cash flow per share | CFO / Shares outstanding | Lower                | Whole payment in CFO, not interest |
| Debt / Equity       | Total debt / Equity      | Slightly lower       | Same debt, more retained earnings  |

## THE ADJUSTMENT YOU ACTUALLY MAKE

To compare an IFRS issuer against a US GAAP peer running operating leases, strip the flat lease expense out of operating costs on the US name and re-class it as amortisation plus interest. Both then show the same EBITDA definition and the same CFO definition, and the multiple you quote means the same thing on each.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, Example 3 question 2 solution, p. 254.*



# Lessor accounting is substantially identical under IFRS and US GAAP — a finance lease derecognises the asset, an operating lease keeps it on the books

*The question to ask about any lessor: is this a financing business (banks, aircraft lessors) or a rental business (real estate)? The statements look nothing alike.*

| Financial statement  | Finance lease (lessor)              | Operating lease (lessor)           |
|----------------------|-------------------------------------|------------------------------------|
| Balance sheet        | De-recognise asset; book receivable | Keep asset; keep depreciating it   |
| Income statement     | Interest revenue, effective rate    | Straight-line lease revenue + depn |
| At inception         | Gain or loss vs carrying amount     | None – this is not a sale          |
| Cash flow statement  | Whole receipt in CFO                | Whole receipt in CFO – identical   |
| Proton machine, Yr 1 | Receivable 316,987; revenue 37,908  | PP&E 280,000; revenue 100,000      |

## THE SAME MACHINE, TWO SETS OF STATEMENTS

Example 4 takes Proton's lessor, whose machine carries at EUR 350,000. Under the finance lease the asset goes and a receivable arrives at the EUR 379,079 present value — assets jump — and Year 1 revenue is EUR 37,908 of interest. Under the operating lease the machine stays, depreciating to EUR 280,000, and Year 1 revenue is the full EUR 100,000. Cash received is EUR 100,000 either way.

Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Lessor Accounting" and Example 4, pp. 254–256.

SECTION 02

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# Pensions & share-based comp

# A defined-contribution plan defines what goes IN and hands every risk to the employee; a defined-benefit plan defines what comes OUT and keeps the risk on the sponsor

*That one sentence is the whole distinction. Everything in the table below is a consequence of it.*

| Dimension          | Defined contribution (DC)        | Defined benefit (DB)                    |
|--------------------|----------------------------------|-----------------------------------------|
| What is promised   | A fixed annual contribution      | A fixed annual pension in retirement    |
| Investment risk    | The employee bears it            | The employer – it must fund shortfalls  |
| Actuarial risk     | The employee                     | The employer – pays to the last retiree |
| P&L expense        | The cash contribution, full stop | Service cost + net interest (IFRS)      |
| Other comp. income | Nothing                          | Remeasurements: actuarial G/L, returns  |
| Balance sheet      | Only an accrued payable          | Net funded status: obligation – assets  |

## READING THE FOOTNOTE

A one-line pension note means defined contribution. Pages of assumptions, sensitivities and funding schedules mean defined benefit, and a legacy workforce. The curriculum illustration is BT Group: a GBP 57,327m obligation against GBP 50,956m of plan assets in 2018, leaving a GBP 6,371m deficit — 29% of all non-current liabilities.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Financial Reporting for Postemployment and Share-Based Compensation Plans", pp. 256–259.*

# The defined-benefit obligation rolls forward from five sources — and only some of them reach profit, which is the entire point of the disclosure

*Separate the recurring from the volatile: service cost recurs every year, actuarial remeasurement is noise that never touches profit under IFRS.*

## ILLUSTRATION · DEFINED-BENEFIT OBLIGATION ROLL-FORWARD · EUR MILLIONS

Opening obligation EUR 800m, discount rate 4.0%. Service cost 35. Past-service cost 5 from a plan amendment. Actuarial loss 15 as the discount rate falls. Benefits paid to retirees 20.

1. Opening obligation at 1 January: 800.0
2. Service cost — PV of the benefit earned by one more year of service: +35.0
3. Interest cost — unwinding the discount,  $800.0 \times 4.0\%$ : +32.0
4. Past-service cost — the benefit formula was enhanced: +5.0
5. Actuarial loss — a remeasurement, driven by the lower discount rate: +15.0
6. Benefits paid to retirees — cash leaving the plan: -20.0
7. Closing obligation:  $800.0 + 35.0 + 32.0 + 5.0 + 15.0 - 20.0 = 867.0$

**IFRS: 35 + 5 service cost and 32 interest hit P&L; the 15 actuarial loss goes to OCI and stays.**

<sup>1</sup> US GAAP sees five components: past-service cost and actuarial gains and losses hit OCI first, then amortise into pension expense over time.

<sup>2</sup> Net interest in profit or loss is charged on the NET liability (obligation less plan assets). This table rolls the gross obligation only.

Source: CFA Institute, Curriculum Vol 4, Learning Module 8, "Accounting for Defined-Benefit Plans under IFRS" and "under US GAAP", pp. 257–259.

# Share-based compensation is fixed at grant-date fair value and expensed over the service period — no cash leaves, but earnings fall and the share count rises

*Grants have overtaken options almost everywhere, for one reason: an RSU is worth something at any share price above zero, while an option can expire worthless.*

## STOCK GRANTS AND RSUs

Fair value at grant = share price × shares granted. Expensed over the service period. No pricing model needed — the value is simply observable on the day.

## STOCK OPTIONS

Fair value must be modelled — Black-Scholes or binomial; no standard prescribes one. Fair value RISES with volatility, a longer expected life and a higher risk-free rate, and FALLS as dividend yield rises.

ILLUSTRATION • 100,000 RSUs • THREE-YEAR SERVICE PERIOD • SHARE AT USD 45

1. 100,000 RSUs granted at a USD 45 share price, vesting after three years of service
2. Grant-date fair value =  $100,000 \times \text{USD } 45 = \text{USD } 4.5\text{m}$  — and that number never changes
3. Expense =  $4.5\text{m} \div 3 = \text{USD } 1.5\text{m}$  a year to SG&A, credited to APIC rather than to cash

USD 4.5m of expense • zero cash out • 100,000 new shares issued at vesting

SECTION 03

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# Income taxes

# Accounting profit follows the accounting standards and taxable income follows the tax code — and every gap between them is either timing or permanent

*Temporary differences reverse; permanent differences never do. That single test decides whether you book deferred tax or simply move the effective rate.*

$$\text{Income tax expense} = \text{Taxes payable} + \Delta \text{ DTL} - \Delta \text{ DTA}$$

*where: Taxes payable follows taxable income. The deferred movements are what drag reported tax expense back onto accounting profit — the matching principle applied to tax.*

| Reconciling item                        | Type         | Effect     | Example                        |
|-----------------------------------------|--------------|------------|--------------------------------|
| Accelerated tax depreciation            | Temporary    | DTL        | Tax 7-yr life vs book 10-yr    |
| Provision for doubtful debts            | Temporary    | DTA        | Deductible only on write-off   |
| Unused tax losses carried forward       | Carryforward | DTA        | Usable against future profits  |
| Non-taxable dividends from a subsidiary | Permanent    | Lowers ETR | Never enters taxable income    |
| Fines, penalties, non-deductible costs  | Permanent    | Raises ETR | Book expense, never deductible |
| Tax credits — R&D, solar, electric cars | Permanent    | Lowers ETR | Cuts the tax bill directly     |

*Source: CFA Institute, Curriculum Vol 4, Learning Module 9, "Differences between Accounting Profit and Taxable Income", pp. 283–287.*

# Compare carrying amount with tax base: an ASSET worth more on the books than on the tax return means tax is still owed — a deferred tax liability

*Four rows, and the liability rows are the exact mirror of the asset rows. Learn the two asset rows properly and the other two follow.*

| Item      | Carrying vs tax base | Gives rise to          | The economic reason                      |
|-----------|----------------------|------------------------|------------------------------------------|
| Asset     | Carrying > tax base  | Deferred tax liability | You will recover more than the tax basis |
| Asset     | Carrying < tax base  | Deferred tax asset     | Tax already paid on more than book value |
| Liability | Carrying > tax base  | Deferred tax asset     | Expensed more than tax has yet allowed   |
| Liability | Carrying < tax base  | Deferred tax liability | Settling it will create taxable income   |

## A DEFERRED TAX ASSET IS ONLY WORTH WHAT YOU CAN USE

A DTA needs future taxable profit to be recovered against. Where that is doubtful, US GAAP raises a valuation allowance writing it down to the amount more likely than not to be realised; IFRS simply reverses it, or never recognises it at all. Same economics, two mechanics. Watch the release: a company turning profitable writes the allowance back and books a one-off credit in tax expense. Strip it out — it is not earnings.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 9, Exhibit 1 "Treatment of Temporary Differences", p. 284.*



# Accelerated tax depreciation is the canonical DTL — it builds while the tax life runs, peaks when the tax base hits zero, then unwinds to nothing

*One asset, two schedules: straight-line over 10 years for the books, straight-line over 7 for the tax authority. The tax authority lends you cash, then takes it back.*

| Item (GBP)                                  | Year 1 | Year 2 | Year 3 |
|---------------------------------------------|--------|--------|--------|
| Carrying amount — book 10-yr life, 2,000/yr | 18,000 | 16,000 | 14,000 |
| Tax base — tax 7-yr life, 2,857/yr          | 17,143 | 14,286 | 11,429 |
| Temporary difference (carrying – tax base)  | 857    | 1,714  | 2,571  |
| Deferred tax liability at a 30% tax rate    | 257    | 514    | 771    |
| Change in DTL, added to income tax expense  | +257   | +257   | +257   |

## READING YEAR 3 • RESTON PARTNERS INCOME TAX EXPENSE

1. Book profit 26,700 but taxable income only 25,843, because tax depreciation is 2,857 against book 2,000
2. Tax actually payable =  $25,843 \times 30\% = 7,753$ , and the DTL rises by  $771 - 514 = +257$
3. Income tax expense =  $7,753 + 257 = 8,010$  — which is exactly 30% of the 26,700 book profit

**A temporary difference changes WHEN tax is paid. It never changes the 30% rate.**

<sup>1</sup> GBP 20,000 of equipment. The DTL peaks at 1,800 in Year 7 when the tax base hits zero, then unwinds to zero by Year 10.

Source: CFA Institute, Curriculum Vol 4, Learning Module 9, Example 3 (Reston Partners), Exhibits 6 to 8, pp. 288–291.

# Three rates, three jobs – statutory is the benchmark, effective drives the earnings forecast, and the cash rate is what actually leaves the building

*Every listed company publishes this walk in its tax note. It is the fastest way to see where a group actually earns its money.*

| Johnson & Johnson – statutory to effective | 2021        | 2020         | 2019         |
|--------------------------------------------|-------------|--------------|--------------|
| US statutory rate                          | 21.0%       | 21.0%        | 21.0%        |
| International operations                   | (16.4)      | (9.9)        | (5.9)        |
| US taxes on international income           | +6.7        | +2.7         | +1.8         |
| Tax benefits from loss on capital assets   | (1.3)       | (1.2)        | (0.3)        |
| Tax benefits on share-based compensation   | (1.0)       | (1.5)        | (0.5)        |
| Tax Cuts and Jobs Act related impacts      | (0.5)       | +0.7         | (3.9)        |
| All other                                  | (0.2)       | (1.0)        | +0.5         |
| <b>Reported effective tax rate</b>         | <b>8.3%</b> | <b>10.8%</b> | <b>12.7%</b> |

One line does almost all of it: international operations, –16.4 points in 2021 alone.

<sup>1</sup> Effective rate = income tax expense ÷ pretax income. JNJ 2021: USD 1,898m ÷ USD 22,776m = 8.3%, against a 21% US statutory rate.

<sup>2</sup> Cash tax rate = cash tax paid ÷ pretax income. It differs from the effective rate only by the period movement in deferred tax.

Source: CFA Institute, Curriculum Vol 4, Learning Module 9, "Corporate Income Tax Rates", Example 5 and Exhibit 15 (Johnson & Johnson), pp. 294–299.

SECTION 04

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# Reporting quality & red flags

# Reporting quality and earnings quality are two different questions — and when reporting quality is low you cannot answer the second one at all

*Reporting quality asks whether the numbers are faithful. Earnings quality asks whether the economics are sustainable. You must settle the first before you can pose the second.*

## LOW financial reporting quality

## HIGH financial reporting quality

**HIGH**  
earnings  
quality

### CANNOT BE ASSESSED

Low reporting quality impedes the assessment of earnings quality, and impedes valuation. The economics may well be excellent — you have no way to establish it.

### THE CELL YOU CAN PRICE

High reporting quality enables the assessment. High earnings quality increases company value. Faithful numbers describing a business that can sustain itself.

**LOW**  
earnings  
quality

### WHERE THE SCANDALS LIVE

Assessment is impeded here too, and the economics are weak underneath. Enron, WorldCom, Wirecard — misleading reports concealing a business that did not work.

### HONEST ABOUT A WEAK YEAR

High reporting quality enables the assessment. Low earnings quality decreases company value. A loss, or profit from non-recurring items, reported faithfully.

<sup>1</sup> Both cells on the left say the same thing: assessment is impeded, so valuation stalls. You are not told earnings are bad, only that you cannot see.

<sup>2</sup> Toyota FY2014 Q1 is bottom right: unit sales fell 1.6% while operating profit rose 87.9% on a weak yen. Faithful reporting, unsustainable earnings.

Source: CFA Institute, Curriculum Vol 4, Learning Module 10, Exhibit 1 "Relationships between Financial Reporting Quality and Earnings Quality", p. 320.

# Reporting quality runs down a six-step spectrum, and the step that matters most is the one where it stops being legal

*Steps 1 to 4 are all within GAAP. The fall from 4 to 5 crosses a legal boundary — above it everything is judgement, below it everything is misstatement.*

- 1 GAAP, decision-useful, sustainable, adequate returns**  
The top. Faithful reporting of a business that earns its cost of capital and can keep doing it.
- 2 GAAP, decision-useful, but sustainable? Low earnings quality**  
The reporting is fine, the economics are not: one-off FX, asset sales, non-recurring items.
- 3 Within GAAP, but biased choices**  
Permitted options picked to flatter — depreciable lives, capitalise versus expense, reserve levels.
- 4 Within GAAP, but earnings management**  
Real EM changes the business to hit a number, such as cutting R&D. Accounting EM stretches estimates.
- 5 Non-compliant accounting**  
Outside GAAP. Misstated revenue, understated liabilities, costs improperly capitalised. Restatement territory.
- 6 Fictitious transactions**  
The floor. Revenue that never happened, counterparties that never existed. Enron, WorldCom, Wirecard, Satyam.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, Exhibit 2 "Quality Spectrum of Financial Reports", p. 321.*

# Three conditions are typically present when low-quality reports are issued — opportunity, motivation and rationalisation — and removing any one of them stops it

*The curriculum is careful here: these are conditions, not causes. Plenty of managers face all three and report honestly anyway.*

## 1 • MOTIVATION

Pressure to hit a number:

- Mask poor performance management believes is temporary
- Meet or beat analyst forecasts and management guidance
- Career concerns — surveyed CFOs rank these ABOVE bonus
- Incentive pay tied to earnings or to the share price
- Avoid breaching a debt covenant

Banking earnings in a strong year is the same instinct running backwards.

## 2 • OPPORTUNITY

The ability to do it undetected:

- Internal — poor internal controls
- Internal — an ineffective or captive board
- External — standards that allow divergent choices
- External — minimal consequences for a bad choice

Disciplining mechanisms push back: securities regulators, the external auditor, private contracting through covenants, and the market pricing the risk into the cost of capital.

## 3 • RATIONALISATION

A justification the person can live with:

- "Everyone in this industry does it"
- "A bridge year — we will true it up next time"
- "The rule is unfair and does not fit our business"
- "I am protecting jobs"

Enron's CFO told a fraud conference he knew it was wrong, so he followed procedure: board approval, legal and accounting opinions, disclosure. Process became the permission.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, "Conditions Conducive to Issuing Low-Quality Financial Reports" and "Mechanisms That Discipline Financial Reporting Quality", pp. 337–339.*

# The warning signs sit in six places, and every one is a ratio you can compute from a published annual report in about ten minutes

*The curriculum's own framing: the choices management makes to hit a number leave a trail, like tracks in sand or snow. These are the tracks.*

| Where to look    | What to compute                   | What deterioration is telling you         |
|------------------|-----------------------------------|-------------------------------------------|
| Revenue          | Growth against peers and industry | Out-of-line growth needs an explanation   |
| Receivables      | DSO; receivables turnover         | Channel stuffing, or fictitious sales     |
| Inventory        | Inventory turnover; days on hand  | Obsolescence not yet written down         |
| Asset allocation | Total asset turnover              | Foreshadows a goodwill write-down         |
| Capitalisation   | Capitalised vs expensed vs peers  | A lone capitaliser is an outlier to check |
| Cash conversion  | CFO ÷ net income as a time series | Below 1.0 or falling: accruals suspect    |

## THE ONE TO BUILD FIRST

Build a time series of cash from operations divided by net income. The curriculum is blunt about the reading: if the ratio is consistently below 1.0, or has declined repeatedly, the company's accrual accounts may have problems. Net income moves the share price; cash pays the bills; sooner or later net income has to arrive as cash or the company does not survive.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 10, "Warning Signs", pp. 372–375.*

SECTION 05

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# Ratio analysis & DuPont



# Four families, four questions – and the family decides whether you use average balances or period-end ones, which is where most computation errors start

*A ratio on its own is a number, not a finding. It needs a history, a peer and an economic story before it means anything at all.*

| Family        | The question it answers       | Three canonical ratios              | Balances used |
|---------------|-------------------------------|-------------------------------------|---------------|
| Activity      | Are my assets working hard?   | Inventory turnover · DSO · Asset TO | Averages      |
| Liquidity     | Can I pay the next 12 months? | Current · Quick · Cash ratio        | Period end    |
| Solvency      | Can I survive the decade?     | Debt/Equity · Debt/Assets · EBIT/I  | Mixed         |
| Profitability | Do I earn on capital?         | Gross · Operating margin · ROE      | Averages      |

## FOUR RULES THAT PREVENT MOST RATIO ERRORS

1. Pair a flow with an average balance. Revenue is earned over a year; assets sit at a date.
2. Liquidity ratios are the exception — they measure a position, so use the closing balance sheet.
3. Annualise before comparing. A quarterly inventory turnover of 3.18 is not comparable to an annual 12.
4. Normalise for leases, share-based comp and restructuring before putting two peers side by side.

<sup>1</sup> The curriculum lists two more under liquidity: the cash conversion cycle (DOH + DSO – days of payables) and the defensive interval ratio.

<sup>2</sup> Context for any ratio: company history, the peer group, the company's own stated goals and strategy, and the phase of the business cycle.

Source: CFA Institute, Curriculum Vol 4, Learning Module 11, "Common Ratio Categories" through "Profitability Ratios", Exhibits 17 to 27, pp. 409–429.

# Liquidity asks whether the company survives the next twelve months and solvency whether it survives the next decade – and a firm can fail either test alone

*Solid Co and Stressed Co are teaching illustrations. The definitions are the curriculum's; the thresholds are market convention, not CFA numbers.*

| Ratio                 | Definition                            | Solid Co | Stressed Co | Convention |
|-----------------------|---------------------------------------|----------|-------------|------------|
| Current ratio         | Current assets / Current liabilities  | 2.4      | 0.9         | > 1.0      |
| Quick ratio           | (Cash + ST investments + AR) / CL     | 1.6      | 0.5         | > 1.0      |
| Cash ratio            | (Cash + ST investments) / CL          | 0.8      | 0.1         | > 0.2      |
| Cash conversion cycle | DOH + DSO – days of payables          | 41 d     | 96 d        | Trend      |
| Debt / Equity         | Total debt / Total equity             | 0.7      | 3.4         | < 2.0      |
| Debt / Assets         | Total debt / Total assets             | 0.30     | 0.65        | < 0.50     |
| Interest coverage     | EBIT / Interest payments              | 8.2      | 1.6         | > 3.0      |
| Fixed-charge coverage | (EBIT + leases) / (Interest + leases) | 5.9      | 1.3         | > 2.0      |

Total debt = interest-bearing short- and long-term debt, per the curriculum. State your definition.

<sup>1</sup> The cash conversion cycle sits under liquidity in the curriculum, not activity — Exhibit 22 lists it as an additional liquidity measure.

<sup>2</sup> Stressed Co fails every line. At 3.4× debt to equity and 1.6× interest coverage, one weak quarter takes the covenant with it.

Source: CFA Institute, Curriculum Vol 4, Learning Module 11, Exhibit 22 "Liquidity Ratios" and Exhibit 25 "Solvency Ratios", pp. 418–423.

## Return on assets is margin times turnover — so two firms can post the same ROA from opposite ends of the business model, and the ratio alone will not say which

*Same 8.0% ROA, opposite routes: the industrial runs 6.00% margin  $\times$  1.33 turnover, while a luxury house would need 10%  $\times$  0.80.*

$$\text{ROA} = \text{Net profit margin} \times \text{Total asset turnover}$$

*where: Then ROE = ROA  $\times$  Leverage, where leverage is average total assets  $\div$  average equity. Return on invested capital, EBIT  $\times$  (1 – tax rate)  $\div$  average debt plus equity, is the version a credit committee reaches for.*

| Firm (illustrative) | Revenue | Net income | Avg assets | Net margin | Asset TO | ROA   |
|---------------------|---------|------------|------------|------------|----------|-------|
| Grocer              | 80,000  | 240        | 10,000     | 0.30%      | 8.00×    | 2.4%  |
| Software            | 10,000  | 2,000      | 8,000      | 20.00%     | 1.25×    | 25.0% |
| Luxury goods        | 6,000   | 900        | 7,500      | 15.00%     | 0.80×    | 12.0% |
| Industrial          | 12,000  | 720        | 9,000      | 6.00%      | 1.33×    | 8.0%  |

### THE COMPARISON RULE

Compare ROA only inside an industry. Every sector sits on its own margin-turnover curve.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 11, Exhibit 27 "Profitability Ratios" and the ROE decomposition, pp. 426–433.*

# The three-way decomposition answers why ROE is what it is — margin, efficiency or leverage — and Anson's answer is that leverage had nothing to do with it

*The fastest diagnostic in analysis: is this ROE move operating (margin, turnover) or financial (leverage)? Three numbers settle it.*

$$\text{ROE} = \text{NI/Revenue} \times \text{Revenue/Avg assets} \times \text{Avg assets/Avg equity}$$

*where: Net profit margin  $\times$  Total asset turnover  $\times$  Financial leverage. The first two multiply to ROA; the third is the only financing term in the chain.*

| Anson Industries | Net profit margin | Total asset turnover | Leverage | ROE      |
|------------------|-------------------|----------------------|----------|----------|
| FY2              | −0.47%            | 0.84×                | 1.60×    | −0.62%   |
| FY3              | 1.13%             | 0.93×                | 1.54×    | 1.62%    |
| FY4              | 1.11%             | 0.95×                | 1.58×    | 1.66%    |
| FY5              | 3.33%             | 1.11×                | 1.60×    | 5.92%    |
| FY2 → FY5        | +380 bps          | +0.27×               | flat     | +654 bps |

<sup>1</sup> Two-way first: ROE = ROA  $\times$  Leverage. Anson FY5 is 3.70%  $\times$  1.60 = 5.92% (Exhibit 32). Split ROA into margin  $\times$  turnover for the three-way.

<sup>2</sup> Every input is an AVERAGE balance — average total assets and average shareholders' equity. Period-end figures give a different, and wrong, answer.

Source: CFA Institute, Curriculum Vol 4, Learning Module 11, "DuPont Analysis", Exhibits 32 and 33 (Anson Industries), pp. 433–434.

# The five-way decomposition pulls tax and interest out of the margin — and it is the version the financial databases actually ship

Use it when the question is "was this operating, financing, or tax?" — because on an earnings call it is almost always one of those three.

$$\text{ROE} = \text{NI/EBT} \times \text{EBT/EBIT} \times \text{EBIT/Rev} \times \text{Rev/Assets} \times \text{Assets/Equity}$$

where: Tax burden  $\times$  Interest burden  $\times$  EBIT margin  $\times$  Total asset turnover  $\times$  Leverage. Tax burden is one minus the average tax rate, so a HIGHER tax burden means a LOWER tax rate — the trap in nearly every exam question on this.

| Amsterdam PLC | Tax burden | Interest burden | EBIT margin | Asset TO | Leverage | ROE    |
|---------------|------------|-----------------|-------------|----------|----------|--------|
| 2014          | 58.96%     | 97.49%          | 13.98%      | 1.44x    | 2.14x    | 24.72% |
| 2015          | 63.12%     | 97.86%          | 13.98%      | 1.47x    | 2.10x    | 26.50% |
| 2016          | 52.10%     | 97.73%          | 11.04%      | 1.71x    | 2.17x    | 20.78% |
| 2017          | 60.50%     | 97.49%          | 7.56%       | 0.99x    | 2.15x    | 9.53%  |

## READING THE 2015 TO 2017 SLIDE

ROE fell from 26.50% to 9.53% in two years. EBIT margin gave up 642 bps and asset turnover 0.48x — between them, the entire move. Interest burden and leverage barely shifted.

Source: CFA Institute, Curriculum Vol 4, Learning Module 11, Exhibit 34 and Example 16 (Amsterdam PLC), pp. 436–437.

SECTION 06

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# 3-statement modeling basics

# A three-statement model is built in one order and only one — revenue down to net income, then the cash flow statement, and the balance sheet last

*The curriculum builds it on Rémy Cointreau, a French spirits group. Cognac was 73% of FY2021 revenue and 94% of operating profit — so this is really a cognac model.*

- 1 Revenue, forecast by segment**  
Volume and price/mix separately. Rémy 2022E: 7% volume and 4% price/mix compound to 11.3% organic growth.
- 2 Operating expenses — COGS then SG&A**  
Held near historical margins or mean-reverted. Segment forecasts checked back against the group total.
- 3 Pro forma EBIT**  
Rémy 2022E EBIT margin 24.8%, rising to 27.6% by 2024E as the pandemic mix effects unwind.
- 4 Non-operating: interest, tax, shares**  
Interest off the debt schedule, tax at a normalised 32% effective rate, diluted share count for EPS.
- 5 Statement of cash flows**  
Start at net income; add D&A and share-based comp; then working capital, capex, buybacks, dividends, debt.
- 6 Balance sheet last — it is linking**  
With the income statement and cash flows done, the balance sheet largely falls out. If it will not tie, the error is upstream.

*Source: CFA Institute, Curriculum Vol 4, Learning Module 12, "Building a Financial Statement Model", Exhibits 2 and 9 (Rémy Cointreau), pp. 455–466.*

# Working capital is forecast in DAYS, not in currency – and in an inventory-heavy business it swings cash flow harder than the margin does

$DOH = 365 \times \text{inventory} \div COGS \cdot DSO = 365 \times \text{receivables} \div \text{revenue} \cdot \text{days of payables} = 365 \times \text{trade payables} \div COGS.$

| Rémy Cointreau (EUR m)             | FY2020     | FY2021     | FY2022E    | FY2023E    | FY2024E    |
|------------------------------------|------------|------------|------------|------------|------------|
| Inventories                        | 1,364      | 1,493      | 1,426      | 1,340      | 1,245      |
| Accounts receivable                | 199        | 158        | 171        | 185        | 200        |
| Accounts payable                   | 534        | 586        | 597        | 604        | 610        |
| Net working capital                | 1,029      | 1,065      | 1,000      | 922        | 835        |
| As a percent of sales              | 100%       | 105%       | 91%        | 78%        | 65%        |
| Days inventory on hand             | 1,431      | 1,650      | 1,500      | 1,350      | 1,200      |
| Days sales outstanding             | 71         | 57         | 57         | 57         | 57         |
| Days payables outstanding          | 561        | 648        | 628        | 608        | 588        |
| <b>Cash from Δ working capital</b> | <b>-56</b> | <b>-36</b> | <b>+64</b> | <b>+79</b> | <b>+87</b> |

<sup>1</sup> Cash conversion cycle = DOH + DSO – days of payables. Rémy FY2021: 1,650 + 57 – 648 = 1,059 days. Cognac must age before anyone drinks it.

<sup>2</sup> Sign convention: a FALL in net working capital is a source of cash, so it prints as a positive on the cash flow statement.

Source: CFA Institute, Curriculum Vol 4, Learning Module 12, "Working Capital Forecasts", Exhibit 11 (Rémy Cointreau), p. 465; cash conversion cycle per Learning Module 11, Exhibit 22, p. 418.



# Scenario analysis is the curriculum remedy for overconfidence – and on Rémy the lowest-growth scenario produces the HIGHEST near-term free cash flow

*Three coherent bundles of assumptions, not three tweaks to one number. Growth, margin and working capital all move together within each row.*

| Scenario       | Organic growth | EBIT margin  | WC % sales | FCFF 22E | FCFF 23E | FCFF 24E |
|----------------|----------------|--------------|------------|----------|----------|----------|
| Base case      | 11.3 → 10.2%   | 24.8 → 27.6% | 91 → 65%   | 230      | 267      | 300      |
| Alt scenario 1 | 4.0% flat      | 23.6% flat   | 86% flat   | 318      | 133      | 129      |
| Alt scenario 2 | As base case   | 25.0% flat   | 90% flat   | 240      | 109      | 105      |

## THE COUNTER-INTUITIVE RESULT • READ THE 2022E COLUMN DOWNWARDS

1. Scenario 1 carries the LOWEST growth and the LOWEST margin of the three — 4.0% and 23.6%
2. Yet its 2022E free cash flow is the HIGHEST of the three: EUR 318m against EUR 230m
3. One line explains it: working capital at 86% of sales rather than 91% releases cash now

Over one year working capital beat both growth and margin. Over three, the base case wins by far.

<sup>1</sup> Studies cited by the curriculum find 90% confidence intervals wrong as often as 40% of the time. Ask where you could be wrong, and by how much.

Source: CFA Institute, Curriculum Vol 4, Learning Module 12, "Overconfidence in Forecasting", Example 1 and Exhibit 15 (Rémy Cointreau), p. 469.

## Deck 8 — FSA III — Taxes, Liabilities, Quality, Ratios & Modeling

- Any ONE of five criteria makes a lease a finance lease. Either way it capitalises: ROU asset = lease liability = PV of payments. Only the lessee income statement differs, and only under US GAAP.

- DC hands every risk to the employee. DB keeps it — the obligation rolls forward from service, interest, past service, actuarial remeasurement and benefits paid, and only some of that reaches profit.

- Statutory, effective and cash tax rates answer different questions. J&J ran an 8.3% effective rate against 21% statutory, and one line — international operations — did almost all of it.

- Anson's ROE rose from -0.62% to 5.92% on margin and turnover with leverage flat. Amsterdam's fell from 26.50% to 9.53% on EBIT margin and turnover, with financing untouched.

- Proton: EUR 500,000 of cash and EUR 500,000 of expense under both classifications. The finance lease front-loads because interest falls on a shrinking balance while amortisation stays flat.

- Carrying amount above tax base on an ASSET means a deferred tax liability. Reston: the DTL builds 257, 514, 771, and income tax expense still lands at exactly 30% of book profit.

- Reporting quality and earnings quality are separate axes. When reporting quality is low you cannot assess the second at all, which is a reason to walk away rather than to apply a discount.

- Build the model in order and forecast working capital in days. On Rémy the lowest-growth scenario delivered the highest 2022E free cash flow — working capital, not margin, drove year one.